International Conference on Computation, Communication, AND Intelligent Systems in Engineering : Submission (664) has been created.

1 message

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Hello,

The following submission has been created.

Track Name: Track 04: Smart Engineering Systems & Sustainable Infrastructure

Paper ID: 664

Paper Title: AI-Assisted EV Range Prediction and Smart Charging Optimization System

Abstract:
Electric vehicle trip planning faces challenges to uncertainty in energy use, battery levels, road conditions and charging needs. This paper introduces an AI-driven system for predicting EV range and optimizing charging. The system combines route planning, machine learning for energy prediction, battery state tracking and charging station selection using mathematical optimization. It works within the Chennai District using browser-based tools for location input, like geolocation address search and map selection. A road network built from OpenStreetMap data enables route calculations. Dijkstra's algorithm finds the path with alternate routes available for users to choose. An XGBoost regression model predicts energy use based on vehicle details, battery condition, road type, speed, weather and driving behavior. The model's output is used with the starting battery state of charge to calculate energy, driving range and arrival SOC ensuring a minimum 25% battery reserve. If the trip exceeds the range the system identifies nearby charging stations that are reachable. It then uses a PuLP-based mixed-integer linear programming model to decide when and where to charge. The system is built with React, MapLibre, FastAPI, MongoDB, Python, OSMnx, NetworkX, XGBoost and PuLP. Testing with 20,000 real EV trip records showed performance: XGBoost achieved an R^2 of 0.9978, a MAE of 0.1481 and RMSE of 0.3613

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Primary Subject Area: Electric Vehicles & Energy Storage Systems

Secondary Subject Areas: Not Entered

Submission Files:
IEEE-paper2.pdf (730 Kb, Wed, 07 Oct 2026 13:02:32 GMT)

Submission Questions Response: Not Entered

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